CYCLOPENTOLATE

Children's refraction
CNS toxicity in children

CHEMICAL NAME 2-(dimethylamino)ethyl (RS)-2-(1-hydroxycyclopentyl)-2-phenylacetate (hydrochloride)
CLASS Antimuscarinic cycloplegic and mydriatic
TRADE NAMES Mydrilate · Cyclogyl · Cyclopentolate Minims

THERAPEUTIC USES / INDICATIONS

- Cycloplegic refraction (especially in children — paralyses accommodation).
- Uveitis/iritis (relieves ciliary spasm, prevents posterior synechiae).
- Fundus examination.

ADMINISTRATION

- **Eye drops:** 1 drop, repeat after 5–10 min; punctal occlusion in children (reduces systemic toxicity).
- Use 0.5% in infants and children <3 months (or avoid in neonates).

DOSAGES

- **Refraction:** 0.5–1% 1 drop, repeated once after 5 min; examine after 30–60 min.
- **Uveitis:** 0.5–1% 1 drop 2–4 times daily.

PHARMACOKINETICS

- **Onset:** 30–60 min. **Duration:** up to 24 h (cycloplegia 6–24 h).

PHARMACODYNAMICS

- Blocks **muscarinic receptors** in ciliary muscle (cycloplegia) and iris sphincter (mydriasis).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Stinging, blurred near vision, photophobia (up to 24 h).
- Dry mouth, flushing.

ADVERSE EFFECTS (SERIOUS)

- CNS toxicity in children: hallucinations, ataxia, restlessness, seizures (especially 2% strength).
- Angle-closure glaucoma.
- Tachycardia, fever.

CONTRAINDICATIONS & CAUTIONS

- Angle-closure glaucoma or narrow angles.
- Down syndrome, spastic paralysis, brain damage in children (↑ sensitivity); caution.
- 2% in children.

KEY INTERACTIONS

- **Other anticholinergics:** additive.
- **Carbachol, pilocarpine:** antagonism.

NURSING MANAGEMENT

- Use lowest effective strength in children; punctal occlusion.
- Observe children for behavioural changes, fever.
- Warn about prolonged blurred vision.

PATIENT EDUCATION

- Vision will be blurred and light-sensitive for up to 24 hours — no driving, cycling; sunglasses help.
- Parents: report confusion, unsteady walking, fever or flushing.

KEY POINTS

- Antimuscarinic cycloplegic; refraction in children.
- CNS toxicity in children — punctal occlusion.
- Lasts up to 24 h.
- Uveitis treatment.